

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Applied business analytics

Practical – III Task

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Date of Assignment:14-08-2026	Date/Time of Submission:14-08-2026

Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Load dataset
df = pd.read_csv("house_price_regression_dataset.csv")

# Print columns to verify
print("Columns in Dataset:", df.columns.tolist())

# Q.1 Correlation between Year_Built and House_Price
corr_q1 = df["Year_Built"].corr(df["House_Price"])
print("\nQ.1 Correlation (Year_Built vs House_Price):", corr_q1)

# Q.2 Full Correlation Matrix
corr_matrix = df.corr(numeric_only=True)
print("\nQ.2 Correlation Matrix:\n", corr_matrix)

# Q.3 & Q.4 Model 1: X = Square_Footage, Y = House_Price
X1 = df[["Square_Footage"]]
y1 = df["House_Price"]
m1 = LinearRegression().fit(X1, y1)
r2_1 = r2_score(y1, m1.predict(X1))

print(f"\nModel 1 (Square_Footage -> House_Price):")
print(f" Coefficient: {m1.coef_[0]:.4f}, Intercept: {m1.intercept_:.4f}")
print(f" R2 Score: {r2_1:.4f}")

# Q.5 Model 2: X = Year_Built, Y = House_Price
X2 = df[["Year_Built"]]
y2 = df["House_Price"]
m2 = LinearRegression().fit(X2, y2)
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```
r2_2 = r2_score(y2, m2.predict(X2))
print(f"\nModel 2 (Year_Built -> House_Price) R2 Score: {r2_2:.4f}")
```

```
# Q.6 Model 3: X = Lot_Size, Y = House_Price
X3 = df[["Lot_Size"]]
y3 = df["House_Price"]
m3 = LinearRegression().fit(X3, y3)
r2_3 = r2_score(y3, m3.predict(X3))
print(f"\nModel 3 (Lot_Size -> House_Price) R2 Score: {r2_3:.4f}")
```

```
# Q.7 Comparison
print("\n=== Comparison of R2 Scores ===")
print(f"Square_Footage -> House_Price: {r2_1:.4f}")
print(f"Year_Built -> House_Price: {r2_2:.4f}")
print(f"Lot_Size -> House_Price: {r2_3:.4f}")
Output:
```

```
Columns in Dataset: ['Square_Footage', 'Num_Bedrooms', 'Num_Bathrooms', 'Year_Built', 'Lot_Size', 'Garage_Size', 'Neighborhood_Quality', 'House_Price']
```

```
Q.1 Correlation (Year_Built vs House_Price): 0.05196736193562004
```

```
Q.2 Correlation Matrix:
```

	Square_Footage	Num_Bedrooms	Num_Bathrooms	Year_Built	\
Square_Footage	1.000000	-0.043564	-0.031584	-0.022392	
Num_Bedrooms	-0.043564	1.000000	0.022848	-0.015820	
Num_Bathrooms	-0.031584	0.022848	1.000000	-0.021063	
Year_Built	-0.022392	-0.015820	-0.021063	1.000000	
Lot_Size	0.089479	-0.009355	0.034923	-0.061050	
Garage_Size	0.030593	0.113761	0.024846	-0.025485	
Neighborhood_Quality	-0.008357	-0.049024	0.017585	-0.009549	
House_Price	0.991261	0.014633	-0.001862	0.051967	

	Lot_Size	Garage_Size	Neighborhood_Quality	House_Price
Square_Footage	0.089479	0.030593	-0.008357	0.991261
Num_Bedrooms	-0.009355	0.113761	-0.049024	0.014633
Num_Bathrooms	0.034923	0.024846	0.017585	-0.001862
Year_Built	-0.061050	-0.025485	-0.009549	0.051967
Lot_Size	1.000000	0.002436	0.037630	0.160412
Garage_Size	0.002436	1.000000	-0.011287	0.052133
Neighborhood_Quality	0.037630	-0.011287	1.000000	-0.007770
House_Price	0.160412	0.052133	-0.007770	1.000000

```
...
```

```
=== Comparison of R2 Scores ===
```

```
Square_Footage -> House_Price: 0.9826
Year_Built -> House_Price: 0.0027
Lot_Size -> House_Price: 0.0257
```